

STAT lecture 1

Basic theory of probability

Disclaimer:

**Please note that these slides are only a brief overview of the topic.
Many more details will be explained on the blackboard and with
MATLAB examples**

Terminology

Stochastic experiment

S = sample space = set of possible outcomes

E = event = combination of "raw" outcomes

Definition of probability (frequency interpretation)

$N_n(E)$ = # outcomes of event E in n independent trials

$$P(E) = \lim_{n \rightarrow \infty} \frac{N_n(E)}{n}$$

Axioms of probability

$$0 \leq P(E) \leq 1, \quad P(S) = 1, \quad P(\emptyset) = 0$$

$$E_1 \cap E_2 = \emptyset \Leftrightarrow P(E_1 \cup E_2) = P(E_1) + P(E_2)$$

Addition rules

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Disjoint set $E_1, \dots, E_n$: $P(\cup_{i=1}^n E_i) = \sum_{i=1}^n P(E_i)$

Complementary event $P(E^c) = 1 - P(E)$

Independent events

A, B independent $\Rightarrow P(A \cap B) = P(A)P(B)$

Conditional probability

Conditional probability $P(A|B) = \frac{P(A \cap B)}{P(B)}$

Multiplication rule $P(A|B)P(B) = P(B|A)P(A) = P(A \cap B)$

Law of total probability:

Disjoint and complete set

$$E_1, \dots, E_n: E_i \cap E_j = \emptyset, \quad \bigcup_{i=1}^n E_i = S$$
$$P(B) = \sum_{i=1}^n P(B|E_i)P(E_i)$$

Bayes rule

$$P(E_j|B) = \frac{P(B|E_j)P(E_j)}{\sum_{i=1}^n P(B|E_i)P(E_i)}$$

Combinatorics

Counting rules often used in probability calculations

Permutations (ordered sequences)

of ordered sequences of n objects: $n! = n*(n-1)*(n-2)*...*2*1$

of ordered sequences of r objects drawn from n : $n!/(n-r)!$

Combinations (unordered sequences)

of unordered sequences of r objects drawn from n : $\binom{n}{r} = \frac{n!}{(n-r)! r!}$